THE DUALIZING COMPLEX AND SHEAF OF A PROPER CURVE BEHAVE CORRECTLY UNDER INTERMEDIATE FIELDS AND BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the adjoint reformulation, the finiteness input, the uniqueness-of-adjoints decomposition, the Hom identification by dimension count, and the base-change derivation of part (2) are all retained verbatim.

Editorial note. The label lemma-sanity-check-duality names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The duality framework being tested. Same chapter, Lemma lemma-duality-dim-1. Fixes omega_X^bullet = a(O_Spec(k)) with a the right adjoint of the twisted inverse image functor, the object whose properties are rechecked.
- Twisted inverse images and their adjoints. Duality for Schemes, Lemma duality-lemma-twisted-inverse-image, Example duality-example-affine-twisted-inverse-image, Lemma duality-lemma-more-base-change and Remark duality-remark-relative-dualizing-complex. Supplies the adjoint decompositions f^! = (f')^! composed with g^!, the affine computation Hom_k(k', k) = k', and the base change theorem for a.
- Finiteness of global functions. Cohomology of Schemes, Lemma coherent-lemma-proper-over-affine-cohomology-finite. Shows k'/k is finite since k' sits inside H^0(X, O_X) of the proper scheme X.

Source reference. Original lemma 02O6: sources/stacks-chow/tags.txt, source lines 3505–3505. Prerequisite pointer invoked at proof line(s) 957; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0A9F: sources/stacks-chow/tags.txt, source lines 12948–12948. Prerequisite pointer invoked at proof line(s) 964; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AA8: sources/stacks-chow/tags.txt, source lines 12976–12976. Prerequisite pointer invoked at proof line(s) 970; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0A9E: sources/stacks-chow/tags.txt, source lines 12947–12947. Prerequisite pointer invoked at proof line(s) 949; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0B6S: sources/stacks-chow/tags.txt, source lines 14080–14080. Prerequisite pointer invoked at proof line(s) 972; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BS2: sources/stacks-curves/pinned-original-source.tex, source lines 796–796. Prerequisite pointer invoked at proof line(s) 929; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a proper scheme of dimension ≤ 1 over a field k. Let $\omega_X^\bullet$ and ω_X be as in Lemma 0BS2.*

(1) *If $X \to \mathrm{Spec}(k)$ factors as $X \to \mathrm{Spec}(k') \to \mathrm{Spec}(k)$ for some field k', then $\omega_X^\bullet$ and ω_X satisfy properties (4), (5), (6) with k replaced with k'.*
(2) *If K/k is a field extension, then the pullback of $\omega_X^\bullet$ and ω_X to the base change X_K are as in Lemma 0BS2 for the morphism $X_K \to \mathrm{Spec}(K)$.*

Proof. Denote $f : X \to \mathrm{Spec}(k)$ the structure morphism. Assertion (1) really means that $\omega_X^\bullet$ and ω_X are as in Lemma 0BS2 for the morphism $f' : X \to \mathrm{Spec}(k')$. In the proof of Lemma 0BS2 we took $\omega_X^\bullet = a(\mathcal{O}_{\mathrm{Spec}(k)})$ where a be is the right adjoint of Duality for Schemes, Lemma 0A9E for f. Thus we have to show $a(\mathcal{O}_{\mathrm{Spec}(k)}) \cong a'(\mathcal{O}_{\mathrm{Spec}(k)})$ where a' be is the right adjoint of Duality for Schemes, Lemma 0A9E for f'. Since $k' \subset H^0(X, \mathcal{O}_X)$ we see that k'/k is a finite extension (Cohomology of Schemes, Lemma 02O6). By uniqueness of adjoints we have $a = a' \circ b$ where b is the right adjoint of Duality for Schemes, Lemma 0A9E for $g : \mathrm{Spec}(k') \to \mathrm{Spec}(k)$. Another way to say this: we have $f^! = (f')^! \circ g^!$. Thus it suffices to show that $\mathrm{Hom}_k(k', k) \cong k'$ as k'-modules, see Duality for Schemes, Example 0A9F. This holds because these are k'-vector spaces of the same dimension (namely dimension 1).

Proof of (2). This holds because we have base change for a by Duality for Schemes, Lemma 0AA8. See discussion in Duality for Schemes, Remark 0B6S. □